

Supreme Counselling for Personal Development

Azure Cloud Infrastructure Proposal

Cloudflare Edge

Azure (Public)

Azure (Internal)

Prepared: June 2026 • Region: US East

Estimated cost: ~BBD \$140–240/mo • Staff access: choose VPN or Cloudflare Zero Trust

1. Executive Summary

This proposal details an Azure-hosted infrastructure for **Supreme Counselling for Personal Development** serving two user groups:

- **Public clients** — website, booking appointments, AI chat assistant
- **Internal staff (<20)** — private RAG system querying company data

The architecture uses **Azure Container Apps** (pay-per-second consumption plan) fronted by **Cloudflare** for DNS, WAF, SSL, and rate limiting. Public and internal data are kept in separate storage — no shared databases, no complex RBAC.

The estimated monthly cost is **~BBD \$140–240** depending on usage, with the primary variable being AI inference load. Staff access to the internal system has **two options** presented in Section 7 for your decision.

Design principles: Separation of concerns • Data sovereignty • Minimal overhead • Pay-per-use scaling • No over-engineering

2. Architecture Overview

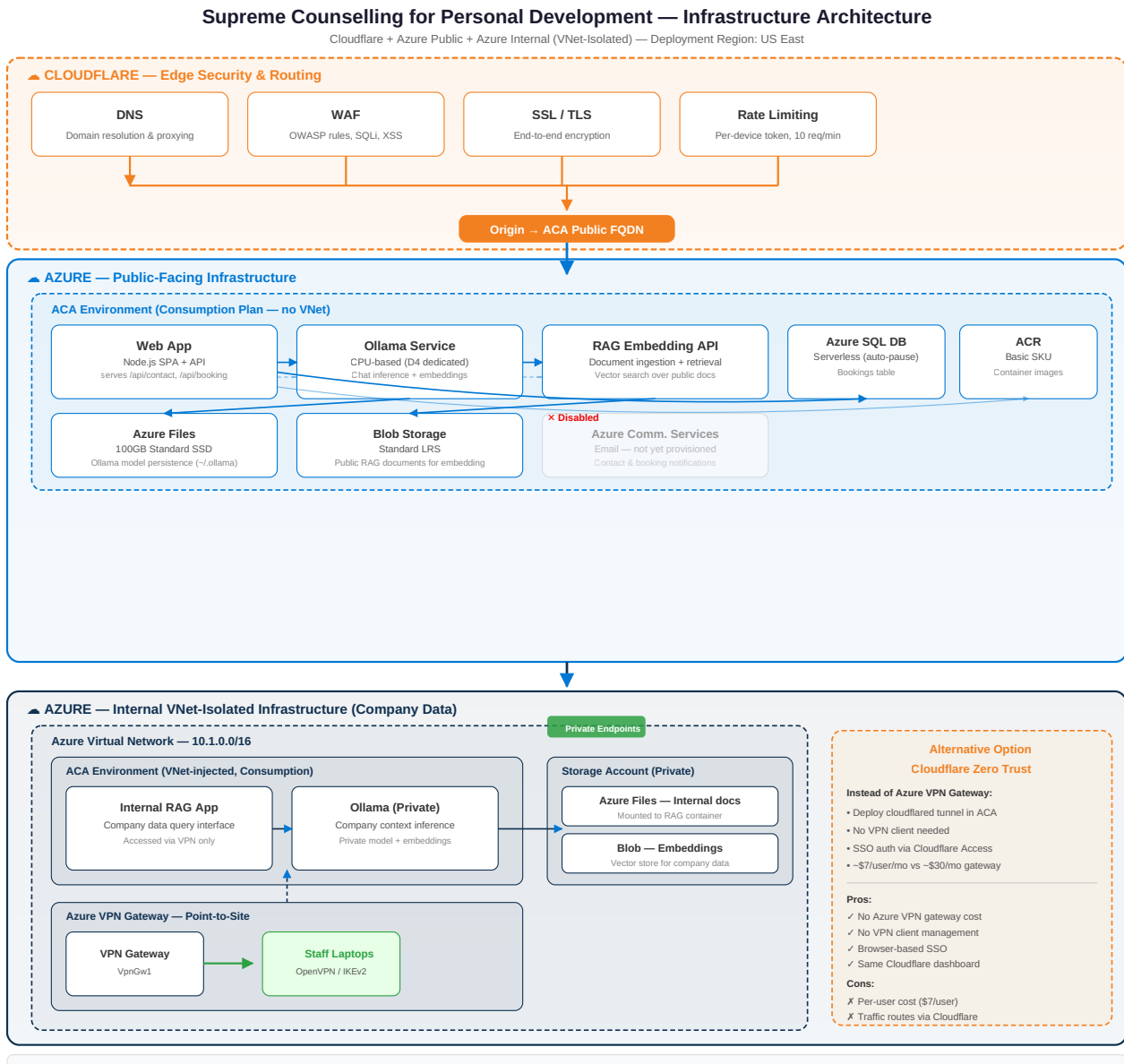


Figure 1: Full infrastructure — Cloudflare → Azure Public → Azure Internal (VNet)

Layer 1 — Cloudflare (Edge) — Free Plan

- **DNS** — proxied record resolution
- **WAF** — OWASP rules, SQLi & XSS protection
- **SSL/TLS** — Full (Strict) end-to-end encryption
- **Rate Limiting** — per-device JWT cookie, 10 req/min on form endpoints
- All public traffic routes through Cloudflare before reaching Azure

Layer 2 — Azure Public (no VNet)

- **ACA Environment (Consumption)** — container orchestration, no VNet needed
- **Container App — Web SPA + API** — Node.js serving frontend and /api/contact, /api/booking
- **Container App — Ollama** — CPU-based chat inference and document embeddings
- **Container App — RAG API** — public document ingestion and vector retrieval
- **Azure SQL Database (Serverless)** — booking records (name, email, phone, date, time)

- **Azure Files** — Ollama model persistence (~/.ollama)
- **Blob Storage** — public RAG documents for embedding
- **Azure Container Registry** — Docker image storage
- **Azure Communication Services** — email notifications (provisioned separately)

Layer 3 — Azure Internal (VNet-Isolated)

- **Azure Virtual Network** (10.1.0.0/16) — fully isolated
- **ACA Environment (VNet-injected)** — containers inside the VNet
- **Container App — Internal RAG** — company data query interface
- **Container App — Ollama (Private)** — private inference on company data
- **Storage Account (Private Endpoints)** — internal company docs + embeddings, traffic never leaves VNet
- **Staff access** — see Section 7 for your two options

3. Component Breakdown

3.1 Cloudflare (Free Plan)

Service	Purpose
DNS	Domain resolution, proxied records
WAF	OWASP Top 10 rules, custom rate limiting
SSL/TLS	Full Strict encryption, automatic cert management
Rate Limiting	10 requests/min per device (JWT cookie) on API endpoints

3.2 Azure Public

Resource	SKU / Config	Purpose
ACA Environment	Consumption (no VNet)	Free container orchestration layer
Container App — Web	Consumption, 0.25 CPU	Node.js SPA + API
Container App — Ollama	Consumption, 1.0 CPU	Chat + embeddings
Container App — RAG API	Consumption, 0.5 CPU	Public document retrieval
Azure SQL Database	Serverless, GP_S_Gen5_1, auto-pause 60min	Bookings
Azure Files	100GB Standard SSD	Ollama model storage
Blob Storage	Standard LRS	RAG documents
Container Registry	Basic SKU	Container images
Log Analytics	Per-GB	Monitoring & logs
Comm. Services	Free tier (placeholder)	Email notifications

3.3 Azure Internal

Resource	SKU / Config	Purpose
Virtual Network	10.1.0.0/16	Isolated network
ACA Environment	VNet-injected	Containers inside VNet
Container App — Internal RAG	Consumption, 0.5 CPU	Company data queries
Container App — Priv. Ollama	Consumption, 0.5 CPU	Private inference
Storage Account	Standard LRS + Private Endpoints	Internal docs + embeddings
Azure Files (Int.)	100GB Standard SSD	Company documents

4. Cost Estimates

All prices in **Barbadian Dollars (BBD)** at the pegged rate: 1 USD = 2 BBD. Based on US East pricing.

Resource	SKU / Detail	BBD/mo
ACA — Web App	Consumption, 0.25 CPU	~\$30
ACA — Ollama (Public)	Consumption, 1.0 CPU	~\$60
ACA — RAG Embedding API	Consumption, 0.5 CPU	~\$30
ACA Env (Public)	Consumption plan	\$0
ACA Env (Internal)	Consumption + VNet injection	~\$20
ACA — Internal RAG + Ollama	Consumption, 0.5 CPU × 2	~\$60
Azure SQL Database	Serverless, 1 vCore, auto-pause	~\$20–60
Azure Files (Ollama)	100GB Standard SSD	~\$24
Azure Files (Internal)	100GB Standard SSD	~\$24
Blob Storage (Public)	Standard LRS, 10GB	~\$2
Blob Storage (Internal)	Standard LRS, 50GB	~\$6
Container Registry	Basic SKU	~\$10
Log Analytics	Per-GB ingestion	~\$10–30
Azure Comm. Services	Free tier (1k emails/mo)	\$0
Cloudflare	Free plan	\$0
Subtotal (infrastructure)		~\$100–160
Staff access	Your choice below	+ \$0 or + \$60
Total		~\$100–220

Real-world note: ACA containers scale to zero when idle. For a small practice with a few dozen visits per day, actual container costs are often **50–70% lower** than the table above — the values shown assume several hours of active use daily. Azure SQL auto-pauses after 60 minutes idle, keeping its cost near zero during off-hours. The realistic monthly bill is likely **~BBD \$140–180** plus the access option.

5. Security & Compliance

Perimeter Security

- All public traffic passes through Cloudflare — DDoS, WAF, SSL
- ACA ingress locked to Cloudflare IP ranges
- No public IP for internal resources

Data Sovereignty

- Public and internal data in separate storage
- No shared database — no RBAC middleware needed
- Internal storage via Private Endpoints — traffic stays in VNet
- Data in US East region

Identity & Access

- Public: JWT cookie for rate limiting (no auth)
- ACA uses managed identity — no connection strings in code
- Secrets stored as ACA secret references
- Internal: VPN or Cloudflare Zero Trust (your choice)

Compliance

- Azure: SOC 2, ISO 27001, HIPAA aligned
- Cloudflare: SOC 2, ISO 27001, PCI DSS
- Option to deploy West Europe for GDPR

6. Scaling Path

Scenario	Action	Cost Impact
Traffic increases	Increase ACA max replicas (3 → 10)	Scales per request, auto-scales down
More staff	Adjust VPN gateway or add Zero Trust users	Minimal
More data	SQL auto-scales; upgrade to Provisioned	~2× if always active
Heavy AI usage	Dedicated CPU profile or GPU VM (NCas T4)	+\$60–700/mo
Need Kubernetes	Migrate ACA → AKS (same containers)	+\$75–150/mo

No code changes needed to scale — ACA abstracts the orchestration. Same containers deploy to AKS if you ever outgrow ACA.

7. Staff Access to Internal System — Two Options

Both options give your staff secure access to the internal RAG system. The difference is **how they connect and how much it costs**.

Option A: Azure P2S VPN

Cost: BBD \$60/mo (fixed)
Setup: Staff install VPN client
Auth: Certificate or Azure AD

Aspect	Detail
Type	Point-to-Site VPN (OpenVPN / IKEv2)
Access level	Full network (L3) — any resource in the VNet
Client needed	Yes — Azure VPN Client app
Always-on cost	BBD \$60/mo regardless of usage
Pros	Full network access No per-user fees Traffic stays on Azure backbone
Cons	Client install & management Fixed cost even when idle Certificate rotation overhead

Option B: Cloudflare Zero Trust

Cost: BBD \$0/mo (Free for up to 50 users)
Setup: Browser-based, no client
Auth: SSO (Google, Microsoft, etc.)

Aspect	Detail
Type	Cloudflare Tunnel + Access (application-layer)
Access level	App-level (browser) — specific URLs only
Client needed	No — any browser
Always-on cost	\$0 on Free plan (up to 50 users)
Pros	No client to manage Browser SSO — staff just log in Free for your team size Same Cloudflare dashboard as rest of stack Built-in audit logs
Cons	App-level access only (not full network) Traffic routes through Cloudflare edge Requires cloudflared sidecar container

Recommendation: For a team of <20, **Cloudflare Zero Trust** is the practical choice — it's free, staff access the internal RAG from any browser with their Google/Microsoft login, no VPN client needed. The only reason to pick the Azure VPN Gateway is if you need full network-level access to the VNet (e.g., staff need to SSH into specific resources, not just hit a web app). The Azure VPN can always be added later if needs change — a simple `enable_vpn_gateway = true` toggle in the OpenTofu config.

8. What's Included in the OpenTofu Plan

Module	Resources	Conditional?
Cloudflare	DNS records, WAF rules, rate limiting rule set	Always
Azure Public	RG, ACA env, 3 container apps, SQL DB, storage, ACR, Log Analytics	Always
Azure Internal	RG, VNet, ACA env (VNet-injected), 2 container apps, storage + private endpoints	Always
VPN Gateway	VPN gateway, local gateway, connection config	Toggle (enable_vpn)
Cloudflare Zero Trust	Tunnel, access application, service token	Toggle (enable_zt)
Comm. Services	Azure Communication Service resource (placeholder)	You provision email key manually

Next steps: Review and pick your staff access option. I'll then finalise the OpenTofu code with the correct toggle, and you can apply it directly to your Azure subscription.

Appendix: Why Not Over-Engineer?

- **No AKS** — ACA provides containers without control plane management
- **No Azure API Management** — the web app routes its own API; rate limiting at Cloudflare
- **No shared database** — public and internal data physically separated, no RBAC middleware
- **No device auth service** — JWT cookies for rate limiting need no database